Forecasting and Macroeconomic Analytics

ECON 522

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Point Forecasts Evaluation

- Good forecasts lead to good decisions.

- Given a track record of forecasts, $y_{t+h,t}$, and corresponding realizations, y_{t+h}, we naturally want to monitor and improve forecast performance.

- We discuss both absolute aspects of forecast evaluation, focusing on methods for checking forecast optimality, and relative aspects, focusing on methods for ranking forecast accuracy, quite apart from optimality.

Review of Point Forecasts

☆ ☆ Exam

$$y_t = \mu + \varepsilon_t + b_1\varepsilon_{t-1} + b_2\varepsilon_{t-2} + \dots$$

$$\varepsilon_t \sim WN\left(0, \sigma^2\right)$$ → is constant

Thus the h-step-ahead linear least-squares forecast is

$$y_{t+h,t} = \mu + b_h\varepsilon_t + b_{h+1}\varepsilon_{t-1} + \dots$$

and the corresponding h-step-ahead forecast error is

errors cancel out

$$e_{t+h,t} = y_{t+h} - y_{t+h,t} = \varepsilon_{t+h} + b_1\varepsilon_{t+h-1} + \dots + b_{h-1}\varepsilon_{t+1}$$

assume stationary σ^2

with variance

$$\sigma_h^2 = \sigma^2\left(1 + \sum_{i=1}^{h-1} b_i^2\right)$$

→ can estimate ~ past information

Standards for Point Forecasts

Property of good forecasts

- The key property of optimal forecast errors is that they should be unforecastable on the basis of information available at the time the forecast was made.
- This unforecastability principle is valid in great generality; it holds, for example, regardless of whether linear-projection optimality or conditional-mean optimality is of interest, regardless of whether the relevant loss function is quadratic, and regardless of whether the series being forecast is stationary.

Review of Forecasts Properties

1. Optimal forecasts are unbiased
2. Optimal forecasts have 1-step-ahead errors that are white noise
3. Optimal forecasts have h-step-ahead errors that are at most $MA(h-1)$
4. Optimal forecasts have h-step-ahead errors with variances that are nondecreasing in h and that converge to the unconditional variance of the process.

you are always more precise in forecasting next month than 10 months from now

How can we check these properties?

- For 1-step ahead, Forecasts errors should be WN and have zero mean: Regress the forecast error on a constant as test that the population mean is zero. DW test is an option
- For multi-steps ahead they will be serially correlated but still have mean zero. Check that they are $MA(h-1)$
- Their variance should non-decreasing in h.

Testing orthogonality of the errors ☆ ☆ Exam

Errors should be orthogonal to the available information. We can test for that with a simple regression

$$e_{t+h,t} = \alpha_0 + \sum \alpha_i x_{it} + u_t.$$

Do an F-test that all the α are jointly insignificant. In particular in

$$e_{t+h,t} = \alpha_0 + \alpha_1 y_{t+h,t} + u_t$$

why should this be zero

optimality corresponds to $(\alpha_0, \alpha_1) = (0, 0)$. This is equivalent to testing the "Mincer-Zarnowitz regression:"

$$y_{t+h} = \beta_0 + \beta_1 y_{t+h,t} + u_t.$$

where optimality implies that $(\beta_0, \beta_1) = (0, 1)$, in which case

$$y_{t+h} = y_{t+h,t} + u_t$$

To get from one to the other substract $y_{t+h,t}$ from each side of the MZ regression.

☆

Accuracy Measures- Forecast Bias

Forecast Bias

How would we rank forecasts?

$$e_{t+h,t} = y_{t+h} - y_{t+h,t}$$

$$\mu_{e_{t+h,t}} = E\left(e_{t+h,t}\right),$$

$$\hat{\mu}_{e_{t+h,t}} = \frac{1}{T}\sum_{t=1}^{T} e_{t+h,t}$$

The mean error is the **forecast bias**.

Accuracy Measures- Error Variance

Error Variance

$$\sigma^2_{e_{t+h,t}} = E\left(e_{t+h,t} - \mu_{e_{t+h,t}}\right)^2,$$

and in sample

$$\hat{\sigma}^2_{e_{t+h,t}} = \frac{1}{T}\sum_{t=1}^{T}\left(e_{t+h,t} - \hat{\mu}_{e_{t+h,t}}\right)^2$$

We want a forecast error with small bias and small variance.

Accuracy Measures- MSE

What if one forecast has smaller bias and larger variance and one the opposite? How would we rank them? Can we take into account both?

$$MSE_{e_{t+h,t}} = E\left(e_{t+h,t}\right)^2$$

and in sample,

$$\boxed{\widehat{MSE}_{e_{t+h,t}} = \frac{1}{T}\sum_{t=1}^{T} e_{t+h,t}^2}$$

measure of accuracy

Sometimes we take the square root to preserve units

$$RMSE_{e_{t+h,t}} = \sqrt{E\left(e_{t+h,t}\right)^2}$$

and

$$\widehat{\text{RMSE}}_{e_{t+h,t}} = \sqrt{\frac{1}{T}\sum_{t=1}^{T} e_{t+h,t}^2}.$$

same unit of forecast

Bias Variance Trade-ff

bias² variance

$$MSE_{e_{t+h,t}} = \sigma^2_{e_{t+h,t}} + \mu^2_{e_{t+h,t}},$$

small MSE is good

and

$$\widehat{MSE}_{e_{t+h,t}} = \hat{\sigma}^2_{e_{t+h,t}} + \hat{\mu}^2_{e_{t+h,t}}.$$

Mean Absolute error is less popular but still used → sum of Absolute value of errors

$$\widehat{MAE} = \frac{1}{T}\sum_{t=1}^{T} |e_{t+h,t}|$$

another option:

Predictive R^2

$$R^2 = 1 - \frac{\sum_{t=1}^{T} e_{t,t-1}^2}{\sum_{t=1}^{T} (y_t - \bar{y})^2}$$ ← variance of y's

Compared with usual R^2, the difference is the numerator.

Predictive R^2 compares an estimate of 1-step-ahead out-of-sample forecast error variance to an estimate of unconditional variance. Put differently, it compares actual 1-step forecast accuracy to that of the historical mean forecast, $\bar{y}$. The hope is that the former is much smaller than the latter, in which case the predictive R^2 will be near 1 .

Theil U-statistics

stick to MSE or RMSE

The so-called "Theil U-statistic" is just a predictive R^2, but we change the benchmark from the historical mean forecast, $\bar{y}$, to a "no change" forecast, y_{t-1}

$$U = 1 - \frac{\sum_{t=1}^{T} e_{t,t-1}^2}{\sum_{t=1}^{T} (y_t - y_{t-1})^2}$$

- In the meteorological literature measures like U are called "skill scores," because they assess actual skill relative to a potentially-naive forecast.
- It is important to note that allegedly-naive benchmarks may not be so naive. For example, many economic variables may in fact be nearly random walks, in which case forecasters will have great difficulty beating the random walk through no fault of their own (i.e., the predictive R^2 relative to a random walk "no-change" forecast given by Theil's U may be near 0)!

Measures of Forecastability

- It is hard to measure the forecastability.
- Example: If I have a WN, the best forecast is zero, yet they two will look very different
- There are limits when looking at predictability
- Never the less we can try to measure it
- TWhether a series is predictable or not should be replaced by one of how predictable it is. Predictability is always a matter of degree.
- How predictable a series is cannot be answered in general. We have to be clear about the relevant forecast horizon and loss function. For example, a series may be highly predictable at short horizons, but not at long horizons.

Predictive Ability

Model A vs Model B

Once we've decided on a loss function, it is often of interest to know whether one forecast is more accurate than another. In hypothesis testing terms, we might want to test the equal accuracy hypothesis,

function of random variables

$$E\left[L\left(e^a_{t+h,t}\right)\right] = E\left[L\left(e^b_{t+h,t}\right)\right]$$

ALF given model B

ALF given model A

against the alternative hypothesis that one or the other is better. Equivalently, we might want to test the hypothesis that the expected loss differential is zero,

low enough

$$E\left(d_t\right) = E\left[L\left(e^a_{t+h,t}\right)\right] - E\left[L\left(e^b_{t+h,t}\right)\right] = \underline{0}$$

ideally 0 (not realistic)

The hypothesis concerns population expected loss; we test it using sample average loss.

Diebod and Mariano Test

The time- t loss differential between forecasts 1 and 2 is then $d_{12t} = L(e_{1t}) - L(e_{2t})$. *DM* requires only that the loss differential be covariance stationary. That is, *DM* assumes that:

$$\text{Assumption } DM: \begin{cases} E(d_{12t}) = \mu, \forall t \\ \text{cov}\left(d_{12t}, d_{12(t-\tau)}\right) = \gamma(\tau), \\ 0 < \text{var}(d_{12t}) = \sigma^2 < \infty \end{cases}$$

The key hypothesis of equal predictive accuracy (i.e., equal expected loss) corresponds to $E(d_{12t}) = 0$, in which case, under the maintained Assumption DM:

$$DM_{12} = \frac{\bar{d}_{12}}{\hat{\sigma}_{\bar{d}_{12}}} \to N(0,1)$$

– distributed normally

where $\bar{d}_{12} = \frac{1}{T}\sum_{t=1}^{T} d_{12t}$ is the sample mean loss differential and $\hat{\sigma}_{\bar{d}_{12}}$ is a consistent estimate of the standard deviation of $\bar{d}_{12}$.

if you don't reject, two models have same predictive ability

Diebold and Mariano Test

- DM is simply an asymptotic z-test of the hypothesis that the mean of a constructed but observed series (the loss differential) is zero. Since the forecast errors, and hence loss differentials, may be serially correlated, the standard error in the denominator of the DM statistic should be calculated robustly.
- A simple approach is to recognize that DM is just a t-statistic for the hypothesis of a zero population mean loss differential, and use HAC SE on an intercept.
- Perhaps an even simpler approach is to regress the loss differential on an intercept, allowing for $AR(p)$ disturbances, and using information criterion like AIC to select p.
- Easy to extend to conditioning variables since can be tested using a simple regression.

Market Based Forecast Combinations

Market Based Forecast Combinations

- Markets can be a spectacularly effective method of information aggregation.
- Hence one might suspect that markets would be useful for combining forecasts.
- Markets and surveys are in certain respects opposite extremes: Markets are loved by economists (as market participants have real money on the line), and surveys are often unloved by economists (as survey participants typically have nothing on the line).
- Both market-based combined forecasts and survey-based combined forecasts are very different from forecasts from a simple single model.

Forward Markets

- Financial markets are naturally forward-looking, and market forecasts can sometimes be extracted from financial asset price.
- A classic example is using forward foreign exchange rates as forecasts of future spot exchange rates.
- Under risk neutrality, agents will always buy forward foreign exchange when it's "cheap" (under-priced) relative to their expectations of the future spot rate, and sell when it's "dear" (over-priced). Immediately then, we have that under risk neutrality:

$$F_t(t+h) = E_t\left(S_{t+h}\right)$$

  where $F_t(t+h)$ is the h-day forward rate prevailing at time t and $E_t\left(S_{t+h}\right)$ is the optimal (conditional mean) spot-rate forecast made at time t for time $t+h$
- Risk Premia, most likely time-varying, are also a possibility.

Futures Market

- Futures markets exist for many many things, trading contracts not only for standard financials (e.g., currencies, interest rates, ...), but also for myriad other things, including aspects of energy, agriculture, metals and other commodities - even weather, real estate, and stock market volatility!
- Because futures are traded on exchanges, you can learn a lot about the contracts traded and currently-prevailing prices (and hence implied forecasts) by visiting exchanges' websites.

Inflation Forecasts from Bond Yields

- Under risk neutrality, nominal government bond yields equal real yields plus expected inflation (the famous "Fisher equation"),

$$i_t(t+h) = r_t(t+h) + E_t\left(\pi_{t+h}\right),$$

  where $i_t(t+h)$ is the nominal bond yield from time t to time $t+h$, $r_t(t+h)$ is the corresponding real yield, and $E_t\left(\pi_{t,t+h}\right)$ the optimal (conditional mean) forecast of inflation between time t and time $t+h$.
- Hence under an assumption about the real rate one can extract expected inflation.

More Markets forecasts

- Bond Yields Forecasts from The Term Premium
- Real Activity Forecasts From the Term Premium
- Real Activity Forecasts From the Default Premium
- Long-Run Equity Return Forecasts from the Dividend Yield
- Volatility Forecasts From Options Markets
- ....

Issues with markets based forecasts

- Market Inefficiencies and No-Arbitrage Conditions
- Moral Hazard and Market Manipulation
- True Moral Issues
- Risk Neutrality

Nice list of website on section 13.5 of FD

Survey Based Forecast Combination

Survey-Based Point Forecast Combination

- A number of groups regularly survey economic and financial forecasters and publish "consensus" forecasts, typically the mean or median - essentially the average! - of the forecasters surveyed. he consensus forecasts often perform very well relative to the individual forecasts.
- The Survey of Professional Forecasters (SPF) is the leading U.S. consensus macroeconomic forecast. It has been produced each quarter since the late 1960s; currently it is produced by the Federal Reserve Bank of Philadelphia. A similar Survey of Professional Forecasters for Europe has been produced each quarter since the late 1990s; it is produced by the European Central Bank.
- Another leading U.S. consensus forecast is the Livingston Survey, which is now also maintained by the Federal Reserve Bank of Philadelphia. It is only bi-annual but has been recorded for more than half a century. There are also many surveys done in the private sector.
- Also Michigan Survey.

Frame Title

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